



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Team: CSE 4104-7A-02

Test Cases

Project Title: AI Sales Analytics Dashboard

Course Name: Software Development III

Course Code: CSE 4104

Submitted to,

Name: Md. Riaz Mahmud,

Assistant Professor, Department of CSE,

Northern University of Business and Technology, Khulna,
Khulna.

1. Test Case Execution Matrix

Test ID	Module	Test Scenario & Input	Expected Result	Status
TC-AUTH-01	Auth	Valid user login (admin@test.com / admin123)	HTTP 200 OK, JWT Bearer token & user payload returned	Passed
TC-AUTH-02	Auth	Login with invalid password	HTTP 401 Unauthorized with error feedback	Passed
TC-AUTH-04	Auth	Login with empty fields	HTTP 422 Unprocessable Content validation error	Passed
TC-AUTH-07	Auth	Admin registration without invite code	HTTP 400 Bad Request, blocks privilege escalation	Passed
TC-AUTH-09	RBAC	Viewer requesting admin endpoint /api/users	HTTP 403 Forbidden with role restriction message	Passed
TC-SALE-01	Sales	Paginated sales list (page=1, size=2)	HTTP 200 OK with exact total and page count metadata	Passed
TC-SALE-02	Sales	Filter sales by category (Furniture)	HTTP 200 OK returning only Furniture records	Passed
TC-SALE-03	Sales	Free-text search (Laptop)	HTTP 200 OK matching product/region names	Passed
TC-SALE-04	Sales	Manager creates sale record	HTTP 201 Created, auto-computed total price	Passed
TC-SALE-05	RBAC	Viewer attempts to create sale	HTTP 403 Forbidden, view-only enforcement	Passed
TC-SALE-08	Sales	Admin deletes sale record	HTTP 200 OK, record safely deleted from DB	Passed
TC-ANL-01	Analytics	Calculate	HTTP 200 OK,	Passed

		Summary KPIs	total revenue, orders, AOV, growth %	
TC-AI-01	AI Engine	Generate AI business insights	HTTP 200 OK, executive summary & recommendations	Passed
TC-AI-02	AI ML	Revenue forecast for next month	HTTP 200 OK, linear regression prediction & confidence bounds	Passed
TC-AI-07	AI Fallback	Gemini 503 / Network spike resilience	Automatic fallback to GroundedAIEngine with zero downtime	Passed

2. Evidence of Test Case Execution

The screenshots are attached below

```

PS C:\myproject\CSE4104_T2\backend> python -m pytest -v
===== test session starts =====
platform win32 -- Python 3.14.4, pytest-9.1.1, pluggy-1.6.0 -- C:\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\myproject\CSE4104_T2\backend
configfile: pytest.ini
testpaths: tests
plugins: anyio-4.14.0
collected 24 items

tests/test_analytics_ai.py::test_analytics_summary_kpis 2026-09-07 05:40:54,501 [INFO] httpx: HTTP Request: GET http://testserver/api/analytics/summary "HTTP/1.1 200 OK"
PASSED
tests/test_analytics_ai.py::test_analytics_overview 2026-09-07 05:40:55,111 [INFO] httpx: HTTP Request: GET http://testserver/api/analytics/overview "HTTP/1.1 200 OK"
PASSED
tests/test_analytics_ai.py::test_ai_insights_generation 2026-09-07 05:41:02,461 [INFO] httpx: HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-flash-latest:generateContent?key=AQ..AB8RN6LdtF-RJ-OsVyf5HM7Ou9qQZRemvAWr2Um6H4BKugAcDg "HTTP/1.1 503 Service Unavailable"
2026-09-07 05:41:02,464 [ERROR] SalesIQ.ai.gemini: Gemini API error 503: {
  "error": {
    "code": 503,
    "message": "This model is currently experiencing high demand. Spikes in demand are usually temporary. Please try again later.",
    "status": "UNAVAILABLE"
  }
}
2026-09-07 05:41:02,471 [INFO] httpx: HTTP Request: POST http://testserver/api/ai/insights "HTTP/1.1 200 OK"
PASSED
tests/test_analytics_ai.py::test_ai_revenue_forecast 2026-09-07 05:41:03,100 [INFO] httpx: HTTP Request: GET http://testserver/api/ai/forecast "HTTP/1.1 200 OK"
PASSED
tests/test_analytics_ai.py::test_ai_anomaly_detection 2026-09-07 05:41:03,706 [INFO] httpx: HTTP Request: GET http://testserver/api/ai/anomalies "HTTP/1.1 200 OK"
PASSED
tests/test_analytics_ai.py::test_ai_chatbot_conversation_greeting 2026-09-07 05:41:12,071 [INFO] httpx: HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-flash-latest:generateContent?key=AQ..AB8RN6LdtF-RJ-OsVyf5HM7Ou9qQZRemvAWr2Um6H4BKugAcDg "HTTP/1.1 503 Service Unavailable"
2026-09-07 05:41:12,074 [ERROR] SalesIQ.ai.gemini: Gemini API error 503: {
  "error": {
    "code": 503,
    "message": "This model is currently experiencing high demand. Spikes in demand are usually temporary. Please try again later.",
    "status": "UNAVAILABLE"
  }
}

```

Screenshot 1

```

2026-09-07 05:41:27,586 [INFO] httpx: HTTP Request: POST http://testserver/api/ai/chat "HTTP/1.1 200 OK"
PASSED
tests/test_auth_rbac.py::test_health_check 2026-09-07 05:41:28,161 [INFO] httpx: HTTP Request: GET http://testserver/api/health "HTTP/1.1 200 OK"
PASSED
tests/test_auth_rbac.py::test_login_success 2026-09-07 05:41:28,977 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/login "HTTP/1.1 200 OK"
PASSED
tests/test_auth_rbac.py::test_login_invalid_password 2026-09-07 05:41:29,871 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/login "HTTP/1.1 401 Unauthorized"
PASSED
tests/test_auth_rbac.py::test_login_nonexistent_user 2026-09-07 05:41:30,469 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/login "HTTP/1.1 401 Unauthorized"
PASSED
tests/test_auth_rbac.py::test_login_empty_fields 2026-09-07 05:41:31,041 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/login "HTTP/1.1 422 Unprocessable Entity"
PASSED
tests/test_auth_rbac.py::test_register_viewer 2026-09-07 05:41:31,865 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/register "HTTP/1.1 200 OK"
PASSED
tests/test_auth_rbac.py::test_register_duplicate_email 2026-09-07 05:41:32,457 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/register "HTTP/1.1 400 Bad Request"
PASSED
tests/test_auth_rbac.py::test_register_admin_without_invite_code 2026-09-07 05:41:33,031 [INFO] httpx: HTTP Request: POST http://testserver/api/auth/register "HTTP/1.1 400 Bad Request"
PASSED
tests/test_auth_rbac.py::test_get_current_user_profile 2026-09-07 05:41:33,638 [INFO] httpx: HTTP Request: GET http://testserver/api/auth/me "HTTP/1.1 200 OK"
PASSED
tests/test_auth_rbac.py::test_unauthorized_access_protected_page 2026-09-07 05:41:34,202 [INFO] httpx: HTTP Request: GET http://testserver/api/auth/me "HTTP/1.1 401 Unauthorized"
PASSED
tests/test_auth_rbac.py::test_role_based_access_admin_only_routes 2026-09-07 05:41:34,800 [INFO] httpx: HTTP Request: GET http://testserver/api/users "HTTP/1.1 403 Forbidden"
2026-09-07 05:41:34,808 [INFO] httpx: HTTP Request: GET http://testserver/api/users "HTTP/1.1 200 OK"
PASSED
tests/test_sales_products.py::test_list_sales_pagination 2026-09-07 05:41:35,403 [INFO] httpx: HTTP Request: GET http://testserver/api/sales?page=1&page_size=2 "HTTP/1.1 200 OK"
PASSED
tests/test_sales_products.py::test_search_and_filter_sales 2026-09-07 05:41:36,165 [INFO] httpx: HTTP Request: GET http://testserver/api/sales?category=Furniture "HTTP/1.1 200 OK"

```

Screenshot 2

```

===== warnings summary =====
..\\..\\Users\\mohaimen\\AppData\\Roaming\\Python\\Python314\\site-packages\\fastapi\\testclient.py:1
C:\\Users\\mohaimen\\AppData\\Roaming\\Python\\Python314\\site-packages\\fastapi\\testclient.py:1: StarletteDeprecationWarning: Using 'httpx' with `starlette.testclient` is deprecated; install 'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa

tests/test_auth_rbac.py::test_login_empty_fields
C:\\Users\\mohaimen\\AppData\\Roaming\\Python\\Python314\\site-packages\\starlette\\exception_handler.py:59: StarletteDeprecationWarning: 'HTTP_422_UNPROCESSABLE_ENTITY' is deprecated. Use 'HTTP_422_UNPROCESSABLE_CONTENT' instead.
  response = await handler(conn, exc) # type: ignore[arg-type]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 24 passed, 2 warnings in 44.43s =====
PS C:\myproject\CSE4104_T2\backend>

```

Screenshot 3